

Oliver Cassidy

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EDUCATION

Imperial College London	London, UK
MEng in Electronic and Information Engineering	Expected Jun 2027
Predicted First-Class Honours; Dean's List 2024; Ranked 2 nd in the year	
Coursework: NN in NumPy, Pipelined ray-tracing engine in Verilog on PYNQ, pipelined RISC-V CPU in SystemVerilog with cache, C90 compiler, Intel FPGA/AWS based game system, WiFi controlled rover, Op-amp from BJT in LTSpice	
The Manchester Grammar School	Manchester, UK
A Levels – Mathematics A*, Further Mathematics A*, Physics A*, Electronics A*; Winner of the Paton Electronics Prize	Jun 2023
Returned by invitation in 2025 to present to sixth form students about research and development in electronics	

PROFESSIONAL EXPERIENCE

Research Intern	London, UK
AMD Research and Advanced Development	Apr 2026-Sep 2026
● Accelerating transformers through FINN using quantization, multi-layer offloading and tiled multiplication, targeting Versal platforms	
● Trained and quantized DeiT, SigLIP, BERT and VLA models across distributed hardware before deploying them with the FINN compiler	
Software Engineering Intern	London, UK
T. Rowe Price	Jun 2025-Sep 2025
● Established a secure research environment on Azure using Terraform and Groovy, deploying OpenAI agents with Bing search and MCP	
● Led a collaboration with Microsoft and global teams to build scalable Terraform and Docker based CI/CD frameworks	
● Implemented in Python a custom MCP server/client with SQL database, agents and Bing search in AI Foundry on a private network	
Private 1:1 Tutoring	Jan 2021-Present
● Used social media to market my own tutoring business and attract clients; Tutored over fifteen students at GCSE and A level	

CONFERENCE PUBLICATIONS

CascadeLUT: Information-Ordered Streaming Inference for Bandwidth-Constrained FPGAs	Ghent, Belgium
Paper published in the International Conference on Field-Programmable Logic and Applications 2026 (FPL)	Jan 2026-Sep 2026
● Implemented a novel statically scheduled cascade architecture with no runtime branching achieving up to 12.5x lower latency	
● Co-designed feature ordering, network topology and FPGA dataflow for streaming deployments, and integrated learned preprocessing	
ReducedLUT: Table Decomposition with “Don’t Care” Conditions	Monterey, USA
Paper published in the ACM/SIGDA International Symposium on Field-Programmable Gate Arrays 2025 (FPGA)	Aug 2024-Feb 2025
● Used C++ for similarity-based table decomposition in ultra-low-latency NNs to reduce area achieving up to 1.63x lower area	

RESEARCH EXPERIENCE

NeuraLUT-Assemble: Hardware-aware Assembling of Sub-Neural Networks for Efficient LUT Inference	Zürich, Switzerland
Tutorial presented at Fast Machine Learning for Science Conference 2025	May 2025-Sep 2025
● Helped deliver LUT-based NN tutorial; Built L-LUT quantization layer and gave live end-to-end PYNQ demo with on-device processing	
Low Latency ML Research	London, UK
Circuits and Systems Research Group, Imperial College London	Jun 2024-Present
● Adapted Python NeuraLUT toolflow to add Verilator testing, CUDA inference, L-LUT compression and Vivado Tcl synthesis	
● Worked with a team from TU Delft to map gate activation and timestep functions to decomposed L-LUTs as part of a brain model	
● Implemented a latency-aware controller for a dynamic NN with early exits and runtime width selection on an ESP32	
● Built a new PPO based DSE solver for fpgaConvNet which matched the SA solver solutions but used only 75% the runtime	
● Wrote a paper on hardware-aware early exits where exits are implemented as Boolean functions in dataflow CNNs and NNs (under review)	

ACADEMIC PROJECTS

Reinforcement Learning Based Racetrack Simulation	May 2026-Jun 2026
● Built RL racing simulator with kinematic bicycle dynamics and vectorized raycasting using Cramer's rule for real-time sensing	
Transformer From Scratch	Jan 2026-Feb 2026
● Implemented Attention Is All You Need in PyTorch, including multi-head attention and trained end-to-end on 4 GPUs with CUDA	
CMATMUL – Cache Based Matrix Multiplication Kernel	Feb 2025-Mar 2025
● Developed an effective C++ based matrix multiplication kernel leading to over a 100x increase in throughput from a naïve solution	
● Optimised the throughput by implementing cache-aware tiling, register blocking, an AVX2 microkernel and OpenMP parallelism	
Collabify	Mar 2024-Aug 2024
● Developed a collaborative web app with personalized music recommendations using a weighted cosine similarity model and Spotify API	
● Designed a production backend with SQL-based persistence, Supabase authentication, Stripe payment integration, and deployed to Render.	
Remote Control Car from Logic	Dec 2022-Mar 2023
● Designed and built a remote control car using a crystal oscillator and filters for a RF pair, logic gates, counters and MOSFET H-bridges	

Technical Proficiencies: Advanced in C++, Python, PyTorch, SystemVerilog, Git; Proficient in C, JavaScript, SQL, Java, Groovy, Verilog, Verilator, React, Basic, Assembly languages, Flask, CUDA, Tcl, Terraform, HTML/CSS

Interests: 3K/5K competitive track running for the Thames Valley Harriers (2023-Present), Raced nationally for junior cycling development teams and trained weekly with British Cycling (2020-2023)